

Figure 10: Case 3

Case 4 (Figure 11)

- You cannot make the learning rate too small or too big.
- If the learning rate is too big (I used 50), there is the possibility of the system becoming unstable, as shown in Figure 11.
- If the learning rate is too small, the learning time will increase

I would like to conclude the article by discussing the computing power needed in building large DNNs, and where the world is headed in this area. The average human brain has 100 billion neurons while one of the artificial DNNs which won the Imagenet competition had 650,000 neurons.

With the increase in the number of neurons, the complexity increases and hence there is a need for more computation. The examples given above train in a few milliseconds. DNNs with millions of neurons and high

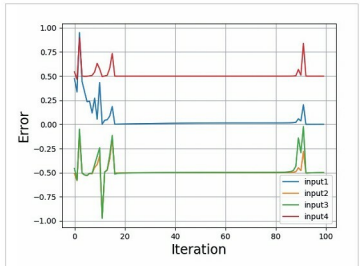


Figure 11: Case 4

training datasets (inputs) take months and years to train a network on a CPU. We can reduce this training time drastically by using GPUs.

Some comparative data is given in Figure 12. Here, cuDNN v2 performance is compared with Caffe using several well-known networks. The CPU is 16-core Intel Haswell E5-2698 2.3GHz with 3.6GHz Turbo. The GPU is NVIDIA GeForce GTX TITAN X.

(The NVIDIA CUDA Deep Neural Network library (cuDNN) is a GPU-accelerated library of primitives for deep neural networks to help increase their performance. The latest cuDNN v4 library yields more performance.)

All major tech companies such as Google, Facebook, IBM, Apple, etc, (the list is too big to cover here) are moving to deep learning nowadays, and using it for applications that range from speech recognition to self-driving cars. 



Figure 12: cuDNN performance comparison with Caffe, using several well known networks (Source: <https://devblogs.nvidia.com>)

References

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- <https://iamtrask.github.io/2015/07/27/python-network-part2/>
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By: Sandip Tambde

The author is currently working at NVIDIA. His interests include automation, deep learning and Python coding. He can be reached at sandip.tambde89@gmail.com.

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